Adwika Vishal Jaiswal

adwikavishal@rediffmail.com · linkedin.com/in/adwikavishaljaiswal · github.com/AdwikaVishal · +91 99999 87746

+91 99999 87746

Education

BMS College of Engineering, Bengaluru 2024 – Present
B.E. in Computer Science CGPA: 9.73

Indraprastha World School (CBSE), New Delhi 2022 – 2024
Class XII: 95.8% Class X: 98.4%

Technical Skills

Languages: Python, Java, C, C++, SQL
AI / ML: Machine Learning, Predictive Modeling, Explainable AI (XAI), OpenCV, MediaPipe, Scikit-learn
Backend & Systems: Spring Boot, REST APIs, Microservices, Data Pipelines
Testing: JUnit, Mockito, MockMvc, H2, JaCoCo, Test-Driven Development
Tools & Platforms: Git, GitHub, Azure, Vercel, CI/CD (Basic)
Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, Probability & Statistics

Experience

Platform Commons May 2025 – Aug 2025
Backend Engineering Intern

- Developed enterprise-grade REST APIs using Spring Boot, enabling structured data exchange across internal services.
- Built comprehensive test suites with JUnit, Mockito, MockMvc, H2, and JaCoCo, improving code coverage and catching regressions early.
- Optimised database queries and backend data flow, improving response reliability and reducing latency for key endpoints.

Humans of EV Mar 2025 – May 2025
Research Intern

- Analysed EV infrastructure adoption trends across urban and semi-urban regions, synthesising findings into structured technical reports.
- Produced data-driven policy briefs used for stakeholder presentations on sustainable mobility systems.

Projects

PresentLoop — AI-Based Real-Time Engagement Detection GitHub

Python, OpenCV, MediaPipe, Scikit-learn | Computer Vision, Machine Learning

- Built a real-time student engagement detection system using OpenCV and MediaPipe for facial and gaze landmark tracking.
- Trained ML classification models on extracted pose and attention features, achieving ∼95% detection accuracy.
- Designed a full end-to-end pipeline from live video input to per-frame prediction and visual dashboard output.

SenseSafe — AI-Powered Emergency Assistance for Persons with Disabilities Drive

Python, Multimodal Sensing | Assistive AI, Real-Time Systems, Human-Computer Interaction

- Built an AI emergency accessibility system for users with visual, auditory, and mobility impairments.
- Designed multimodal sensing algorithms for real-time emergency detection, automated SOS alerts, and adaptive navigation.
- Applied inclusive AI principles to ensure reliability in low-connectivity and real-world deployment environments.

FinCoach AI — Behavioural Finance and Risk-Aware Budgeting Agent

Python, Scikit-learn | Predictive Modeling, Explainable AI, Behavioural Data Analysis

- Built an ML forecasting pipeline on behavioural spending data to detect early indicators of financial stress.
- Improved 30-day budgeting prediction accuracy by 75% over a baseline rule-based model.
- Applied explainable decision modeling (feature importance, SHAP-style reasoning) to surface actionable budget insights.

AI Credit Score Analysis Platform — Explainable ML for Inclusive Lending — Live

Python, XAI Techniques, Vercel | Machine Learning, Algorithmic Fairness — Smart India Hackathon Finalist

- Developed an explainable ML credit scoring system using alternative data sources to reduce bias against underbanked applicants.
- Integrated XAI techniques (feature attribution, decision transparency) enabling loan officers to audit model reasoning.
- Validated platform across 20+ real loan profiles, confirming fairness and consistency across demographic groups.

ReliefLink — Distributed Resource Allocation for Disaster Response

Python, Distributed Systems | Optimisation, AI for Social Good

- Built a distributed AI system for real-time disaster response coordination across multiple agencies.
- Designed intelligent volunteer-to-need matching algorithms, reducing average response latency by 45%.
- Implemented fault-tolerant architecture ensuring system resilience under network failures and partial outages.

PharmaOrchestrator — Agent-Based Workflow Orchestration for Healthcare — Live

Spring Boot, REST APIs, Vercel | Multi-Agent Systems, Workflow Scheduling

- Built an agent-based orchestration system managing multi-step healthcare workflows including patient triage and resource allocation.
- Designed REST APIs for dynamic task scheduling and AI-driven execution pipelines across distributed workflow stages.
- Used real patient dataset (health.csv) for prioritisation modeling, improving simulated throughput and scheduling efficiency.

PhishGuard Nexus — Unified Risk Intelligence Platform — AWS Builder

Python, AWS, Machine Learning | Cybersecurity, Risk Intelligence

- Built a unified risk intelligence platform integrating multiple signals to detect phishing, fraud, and malicious network activity.
- Implemented ML-driven threat classification and real-time risk scoring to support security decision-making.
- Designed signal fusion architecture combining URL analysis, behavioural patterns, and metadata for higher detection precision.

Achievements

- **1st Place** — 404: Skills Not Found Hackathon
- **1st Place** — Contribution Sprint
- **Smart India Hackathon Finalist** — National-level government innovation competition
- **HackHatch, IIIT Delhi** — Top 25 Finalist
- **Winter of Code** — Top 10 Contributor (open source)
- **INSPIRE Award** — Government of India (awarded to top science innovators at school level)

Relevant Coursework

Data Structures & Algorithms, Discrete Mathematics, Probability & Statistics, Operating Systems, Computer Networks, Computer Architecture

Research Interests

Algorithms and Data Structures, Machine Learning, Explainable AI, Human-Computer Interaction, Optimisation, Assistive Technology, Natural Language Understanding

Additional

- Exploring the Salesforce ecosystem (Trailhead): CRM fundamentals, Apex basics, SOQL.
- Author of *The Mystery of Bermuda Triangle* (Kindle) and co-author of *Resilient Rainbow* (poetry anthology).